CS302 Operating Systems

Unit 2: CPU Scheduling

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1. Why scheduling matters

The CPU scheduler picks which ready process runs next. Goals: high CPU utilisation, high throughput, and low waiting, turnaround and response times.

2. First-Come, First-Served (FCFS)

Processes run in arrival order. Simple, but a long job delays all short ones behind it (the convoy effect), so average waiting time can be high.

3. Shortest Job First (SJF)

Runs the process with the shortest next CPU burst. It gives the minimum average waiting time for a given set of processes, but needs burst lengths in advance and can starve long jobs.

4. Round Robin (RR)

Each process gets a fixed time quantum, then goes to the back of the ready queue. Good response time for interactive systems. Too small a quantum adds context-switch overhead; too large and RR behaves like FCFS.

5. Worked example

P1 = 8, P2 = 4, P3 = 2 (all arrive at 0). FCFS average wait = 6.67; SJF average wait = 2.67; RR (quantum 2) average wait = 5.33. Draw a Gantt chart for each to check your answer.

6. Key terms

Waiting time = turnaround time - burst time. Turnaround time = completion time - arrival time. Preemptive schedulers (RR, preemptive SJF) can interrupt a running process.